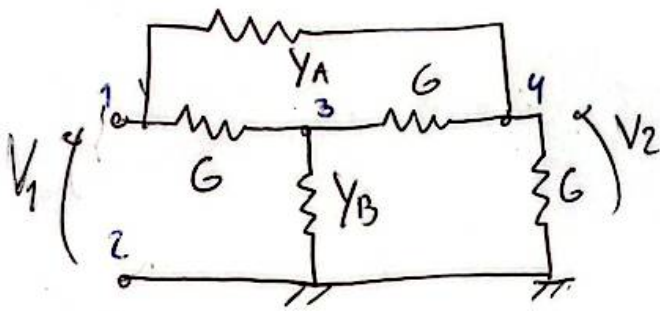


1)



$$Y_A \cdot Y_B = G^2$$

$$Y_{MAT} = \begin{pmatrix} G+Y_A & 0 & -G & -Y_A \\ 0 & Y_B+G & -Y_B & -G \\ -G & -Y_B & 2G+Y_B & -G \\ -Y_A & -G & -G & 2G+Y_A \end{pmatrix}$$

$$Z_{in} = Z_{11} = \frac{Y_{12}^2}{Y_{11}} = \frac{(2G+Y_B)(2G+Y_A) - (-G) \cdot (-G)}{(G+Y_A) \cdot [(G+Y_B)(2G+Y_B) - (-Y_B) \cdot (-Y_B)] + (-G) \cdot [(-G) \cdot (Y_B+G)]}$$

$$Z_{12} = \frac{4G^2 + 2 \cdot G \cdot Y_A + 2 \cdot G \cdot Y_B + Y_B \cdot Y_A - G^2}{G^3 + 2 \cdot G^2 \cdot Y_B + 2G^2 \cdot Y_A + 3G \cdot Y_A \cdot Y_B}$$

$$Y_A \cdot Y_B = G^2$$

$$Z_{12} = \frac{4G^2 + 2G \cdot Y_A + 2G \cdot Y_B}{G \cdot (4G^2 + 2G \cdot Y_A + 2G \cdot Y_B)} \rightarrow Z_{12} = \frac{1}{G} = Z_{in} = Z_{11}$$

$$A_{12}^{42} = \underbrace{Z_{12}^{42}}_{Z_{12}} \cdot \frac{1}{Z_{12}} = \frac{1}{G}$$

$$Z_{12}^{42} = \frac{Y_{42}^{12}}{\frac{Y_{42}^4}{G^4}} = \frac{(-G)(-G) + (-Y_A) \cdot (2G + Y_B)}{G^3 + 2 \cdot G^2 \cdot Y_B + 2 \cdot G^2 \cdot Y_A + 3 \cdot G \cdot Y_A Y_B}$$

$$Z_{12}^{42} = \frac{G^2 + 2GY_A + Y_A + Y_B}{G^3 + 2G^2Y_B + 2G^2Y_A + 3GY_AY_B}$$

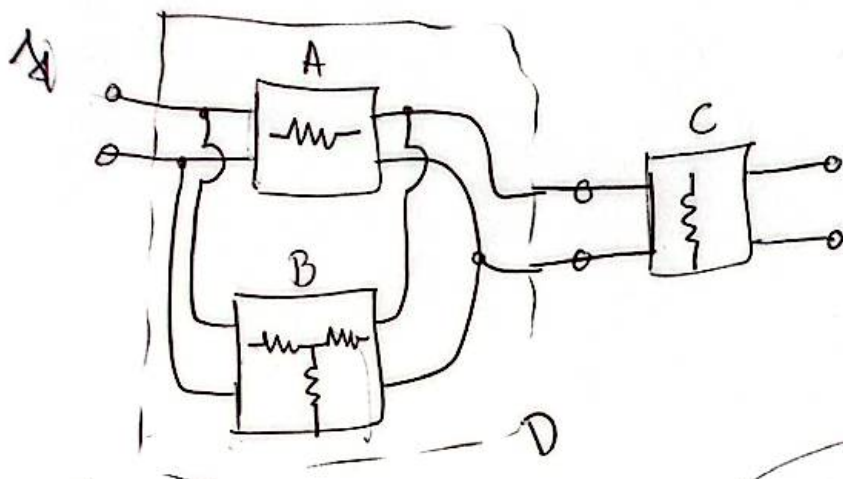
$$\text{si } Y_A Y_B = G^2$$

$$Z_{12}^{42} = \frac{G^2 + 2G \cdot Y_A + G^2}{G^3 + 2G^2Y_B + 2G^2Y_A + 3G^3} = \frac{2G^2 + 2GY_A}{4G^3 + 2G^2Y_B + 2GY_A \cdot G^2}$$

$$Z_{12}^{42} = \frac{2G + Y_A}{4G^2 + 2G \cdot (Y_A + Y_B)}$$

$$A_{12}^{42} = \frac{2G + Y_A}{4G^2 + 2G \cdot (Y_A + Y_B)} \cdot G \rightarrow A_{12}^{42} = \frac{2G + Y_A}{4G + 2 \cdot (Y_A + Y_B)} = \frac{V_2}{V_1}$$

TPC 7
3



~~By~~ $\Delta z = \begin{pmatrix} 1 & z_A \\ 0 & 1 \end{pmatrix}$ $\rightarrow A_y = \begin{pmatrix} 1 & -\frac{Y_A^{-1}}{z_A} \\ 0 & 1 \end{pmatrix}$

~~By~~ $Y_{11} = \frac{I_1}{V_1} \Big|_{V_2=0}$ $\rightarrow Y_{11} = Y_A \parallel (Y_B + Y_C)$

~~$Y_{12} = \frac{I_1}{V_2} \Big|_{V_1=0}$~~

$Y_{22} = \frac{I_2}{V_2} \Big|_{V_1=0}$ $\rightarrow Y_{22} = Y_{11}$ $\rightarrow$ este caso

~~$Y_{12} = \frac{I_1}{V_2} \Big|_{V_1=0}$~~ $\rightarrow Y_{12} = \frac{I_1}{V_2} \Big|_{V_1=0} \rightarrow \frac{I_1 - I_2 \cdot Y_C}{Y_A + Y_B}$

$Y_{12} = -\frac{Y_C}{Y_A + Y_B} \cdot \left(\frac{I_2}{V_2} \right) \rightarrow Y_{12} = -\frac{Y_C}{Y_A + Y_B} \cdot \frac{Y_C \parallel (Y_A + Y_B)}{Y_A + Y_B}$

$Y_{21} = \frac{I_2}{V_1} \Big|_{V_2=0}$ $\rightarrow$ este caso $\rightarrow Y_{21} = Y_{12}$

$$B_Y = \begin{pmatrix} Y_R / (Y_B + Y_R) & -\frac{Y_R}{Y_R + Y_B} \cdot Y_R / (Y_B + Y_R) \\ -\frac{Y_R}{Y_R + Y_B} \cdot Y_R / (Y_B + Y_R) & Y_R / (Y_B + Y_R) \end{pmatrix}$$

A_Y y B_Y están en paralelo $\therefore D_Y = A_Y + B_Y$

$$D_Y = \begin{pmatrix} \left[Y_R / (Y_B + Y_R) \right] + 1 & -\frac{1}{Y_A} - \frac{Y_R}{Y_R + Y_B} \left[Y_R / (Y_B + Y_R) \right] \\ -\frac{Y_R}{Y_R + Y_B} \cdot Y_R / (Y_B + Y_R) & \left[Y_R / (Y_B + Y_R) \right] + 1 \end{pmatrix}$$

$$\Delta C_Y = \begin{pmatrix} 1 & 0 \\ Y_R & 1 \end{pmatrix} \rightarrow \Delta C_Y = -\frac{1}{Y_R} \cdot \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

$$\Delta Y_C = 1$$

TP27
5

$$\Delta Y_D = \left[\left(\frac{Y_R}{Y_B + Y_R} \right) + 1 \right]^2 - \frac{Y_R}{Y_R + Y_B} \cdot \left[\frac{Y_R}{Y_B + Y_R} \right] \cdot \left[\frac{1}{Y_A} + \frac{Y_R}{Y_R + Y_B} \cdot \left(\frac{Y_R}{Y_B + Y_R} \right) \right]$$

$$\Delta Y_D = \left[\left(\frac{Y_R}{Y_B + Y_R} \right) + 1 \right]^2 - \frac{Y_R}{Y_R + Y_B} \cdot \left[\frac{Y_R}{Y_B + Y_R} \right] - \frac{Y_R}{Y_A \cdot (Y_R + Y_B)} \cdot \left[\frac{Y_R}{Y_B + Y_R} \right]$$

$$T_{YD} = - \frac{1}{\frac{Y_R}{Y_R + Y_B} \cdot \left[\frac{Y_R}{Y_B + Y_R} \right]} \cdot \left[\begin{array}{cc} \left[\frac{Y_R}{Y_B + Y_R} \right] + 1 & 1 \\ \Delta Y_D & \left[\frac{Y_R}{Y_B + Y_R} \right] + 1 \end{array} \right]$$

$$D \text{ y } C \text{ están en cascada} \rightarrow T_{YD} \cdot T_C = T_{\text{total}}$$

PL 7
6

$$T_{\text{total}} \approx \frac{1}{Y_R} \cdot \frac{1}{\frac{Y_R}{Y_R + Y_B} \cdot \left[\frac{Y_R}{Y_B + Y_R} \right]}$$

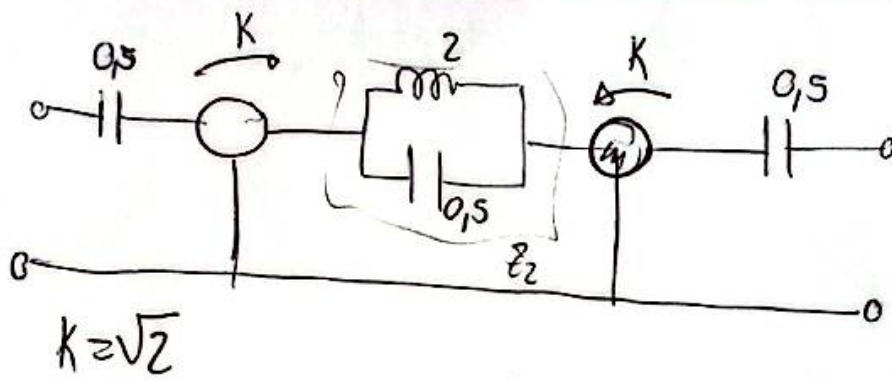
$$\left[\frac{Y_R}{Y_B + Y_R} \right] + 2$$

$$\left[\frac{Y_R}{Y_B + Y_R} \right] + 2$$

$$\Delta y_D + \left[\frac{Y_R}{Y_B + Y_R} \right] + 1$$

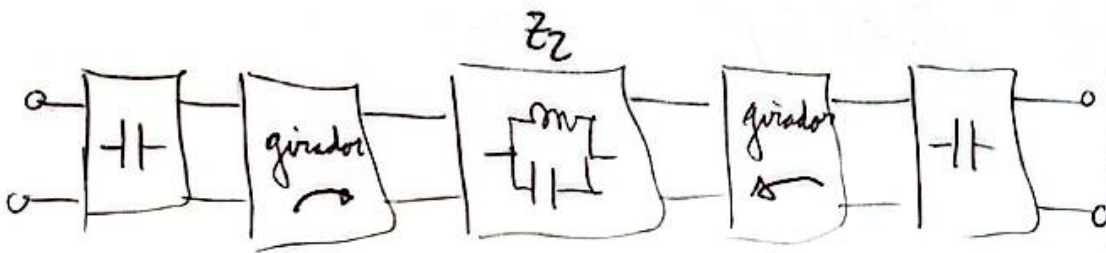
$$\Delta y_D + \left[\frac{Y_R}{Y_B + Y_R} \right] + 1$$

4/2/20



TP7.2
1

$$Z_{\text{Lattice}} = \begin{pmatrix} \frac{Z_A + Z_B}{2} & \frac{Z_B - Z_A}{2} \\ \frac{Z_B - Z_A}{2} & \frac{Z_A + Z_B}{2} \end{pmatrix}$$



$$T_C = \begin{pmatrix} 1 & \frac{1}{sC} \\ 0 & 1 \end{pmatrix}$$

$$Z_2 = \frac{sL \cdot \frac{1}{sC}}{sL + \frac{1}{sC}} = \frac{\frac{L}{s}}{\frac{s^2 LC + 1}{sC}} \Rightarrow \frac{sC}{LC(s^2 + \frac{1}{LC})} \Rightarrow Z_2 = \frac{s \cdot \frac{1}{C}}{s^2 + \frac{1}{LC}}$$

$$T_{Z_2} = \begin{pmatrix} 1 & Z_2 \\ 0 & 1 \end{pmatrix}$$

$$T_{\text{gyrator}} = \begin{pmatrix} 0 & K^2 \\ \frac{1}{K^2} & 0 \end{pmatrix}$$

HP 7.2
2

$$T_1 = T_{gin} \cdot T_c \Rightarrow T_1 = \begin{pmatrix} 0 & K^2 \\ \frac{1}{K^2} & \frac{1}{K^2 \cdot SC} \end{pmatrix}$$

$$T_2 = T_{z_2} \cdot T_1 \rightarrow T_2 = \begin{pmatrix} \frac{z_2}{K^2} & K^2 + \frac{z_2}{K^2 \cdot SC} \\ \frac{1}{K^2} & \frac{1}{K^2 \cdot SC} \end{pmatrix}$$

$$T_3 = T_{gin} \cdot T_2 \rightarrow T_3 = \begin{pmatrix} 1 & \frac{1}{SC} \\ \frac{z_2}{K^4} & 1 + \frac{z_2}{K^4 \cdot SC} \end{pmatrix}$$

$$T_{total} = T_c \cdot T_3 \rightarrow T_{total} = \begin{pmatrix} 1 + \frac{z_2}{K^4 \cdot SC} & \frac{1}{SC} + \frac{1}{SC} + \frac{z_2}{K^4 \cdot (SC)^2} \\ \frac{z_2}{K^4} & 1 + \frac{z_2}{K^4 \cdot SC} \end{pmatrix}$$

TP/L7.2
3

$$\Delta z_{\text{lattice}} = \begin{pmatrix} \frac{z_A + z_B}{2} & \frac{z_B - z_A}{2} \\ \frac{z_B - z_A}{2} & \frac{z_A + z_B}{2} \end{pmatrix} \rightarrow \Delta z = \frac{(z_A + z_B)^2}{4} - \frac{(z_B - z_A)^2}{4}$$

$$T_{\text{lattice}} = \begin{pmatrix} \frac{z_A + z_B}{z_B - z_A} & \frac{1}{2} \left[(z_A + z_B) - \frac{(z_B - z_A)^2}{z_A + z_B} \right] \\ \frac{2}{z_B - z_A} & \frac{z_A + z_B}{z_B - z_A} \end{pmatrix}$$

$$T_{\text{total}} = T_{\text{lattice}}$$

$$1 + \frac{z_2}{K^4 \cdot SC} = \frac{z_A + z_B}{z_B - z_A} \rightarrow \frac{K^4 \cdot SC}{K^4 \cdot SC + z_2} = \frac{z_B - z_A}{z_A + z_B} \quad (1)$$

$$\frac{z_2}{K^4} = \frac{2}{z_B - z_A} \rightarrow z_B - z_A = \frac{2 \cdot K^4}{z_2}$$

$$z_B = \frac{2K^4}{z_2} + z_A \quad (2)$$

$$\frac{2}{SC} + \frac{z_2}{K^4 \cdot (SC)^2} = \frac{1}{2} \cdot \left[(z_A + z_B) - \frac{(z_B - z_A)^2}{z_A + z_B} \right] \quad (3)$$

PL 7.2
4

reemplazo (1), (2) en (3)

$$\frac{4 \cdot K^4 (sC)^4 + z_2 \cdot z}{K^4 (sC)^2} = z_A + \frac{z K^4}{z_2} + z_A + \frac{K^4 sC}{K^4 sC + z_2} \cdot \left(\frac{z K^4}{z_2} + z_A - z_A \right)$$

$$z \cdot z_A = \frac{z^2 K^4 sC + z_2 \cdot z}{K^4 (sC)^2} - \frac{z K^4}{z_2} - \frac{K^4 sC \cdot z}{z_2 \cdot (K^4 sC + z_2)}$$

$$z_A = \frac{\left(\frac{z^2 K^4 sC + z_2 \cdot z}{K^4 (sC)^2} \cdot z_2 \cdot (K^4 sC + z_2) - K^4 \cdot K^4 (sC)^2 \cdot (K^4 sC + z_2) - K^8 sC \cdot K^4 (sC)^2 \right)}{K^4 (sC)^2 \cdot z_2 \cdot (K^4 sC + z_2)}$$

$$z_A = \frac{z K^8 (sC)^2 z_2 + z K^4 sC z_2^2 + K^4 sC z_2^2 + z_2^3 - K^{12} (sC)^3 - K^8 (sC)^2 z_2 - K^{12} (sC)^3}{K^8 (sC)^3 z_2 + K^4 (sC)^2 z_2^2}$$

$$z_B = \frac{z K^4}{z_2} + z_A$$